KadaiMate: Problem Statement & Build Plan

Oct 30, 2026 · @Deepak

Problem statement

KadaiMate is a Tanglish voice ledger for small shops: the shopkeeper speaks, and the app records credit (udhaar), payments, sales and stock, then confirms aloud in Tamil.

Small shops in Tamil Nadu run on a paper notebook or memory. Forgotten or wrong udhaar entries cost real money, and asking a regular customer to pay is awkward. Billing apps assume typing and English.

Shopkeepers talk in code-mixed Tamil and English, use Tamil number words (nooru, aimbadhu, onnara, kaal, arai) and call customers by nicknames and honorifics (Ravi anna, Kutty). The shop is loud. Existing voice ledgers handle Hinglish, not this.

What we build: a phone-first web app where the shopkeeper presses a button and speaks. Gnani Prisma transcribes the speech, our parser turns it into ledger actions, and Gnani Timbre speaks the result back in Tamil.

Success by Nov 10: a working app that handles real Tanglish voices in a real shop setting, a measured accuracy table, and a 60-second demo video posted on LinkedIn and X.

Users, and what already exists

Primary users are kirana owners, tea and vegetable stalls, and small wholesale dealers in Tamil Nadu towns. The demo user is one shopkeeper and one regular customer.

KiranaConnect, another entry, already does Hinglish voice in and out, multi-intent parsing, a spoken "haan" confirmation, stock updates, supplier orders, pending-dues lookup and a dashboard. It reports 13/16 commands clean and 11/16 on noisy phone audio, tested only on synthetic voices (figures from its post, as summarised by our team).

Area	KiranaConnect	KadaiMate adds
Language	Hinglish / Hindi	Tanglish: Tamil number words and fractions, honorifics stripped, phonetic name matching, voice-taught aliases
Udhaar	Reads pending dues	Full ledger: credit, part payments, spoken correction ("205 illa 250"), undo, aging, credit-limit warning

Area	KiranaConnect	KadaiMate adds
Noise handling	Logs misses	Asks to confirm only on low confidence or an unusual amount; reads big amounts digit by digit
Recovery	None	Polite Tamil voice-note reminder sent to the customer
Proof	Dashboard	Audio clip stored with every entry, for disputes
Testing	Synthetic voices	Real-voice accuracy table: real speakers, clean / shop noise / phone

Our edge in one line: built for how Tamil shopkeepers actually talk, and tested on real voices.

Scope

We build the Must list first and stop adding features on Nov 6, whatever is left. Each item lists an example the shopkeeper says.

Must (done by Nov 4)

1. Push-to-talk capture on a phone browser, transcript shown on screen.
2. Tanglish parser: Tamil number words and fractions (nooru aimbadhu = 150, onnara = 1.5, kaal = 0.25, arai = 0.5), honorifics stripped (Ravi anna = Ravi), several actions in one sentence.
3. Udhaar ledger: add credit, record full or part payment, ask balance. "Ravi anna ku 250 kadan, rice and oil" / "Ravi 100 kuduthaar" / "Ravi ku evlo baaki?"
4. Spoken correction and undo: "205 illa 250", "adha cancel pannu".
5. Smart confirmation: confirm only when transcription confidence is low or the amount is unusual for that customer; read large amounts digit by digit.
6. Phonetic name matching against the customer list, with a question when two names match ("Ravi Kumar-a, Ravi Shankar-a?").
7. Timbre reply in Tamil, short by default: "Ravi ku 250 add pannitten. Total 600."
8. Daily sales total and a spoken closing-time summary: "Inniku evlo vyaapaaram?"
9. Live dashboard: customers, balances, today's entries.

Should (Nov 5–6)

- Polite reminder: "Ravi ku reminder anuppu" creates a Tamil voice note with Timbre, shared to the customer through WhatsApp's share sheet.
- Aging: "30 naal ku mela baaki yaaru?" and a credit-limit warning.

- Audio clip saved with each entry, playable as proof.
- Voice-taught aliases: "Kutty na Karthik".
- Stock in and low-stock query: "Sugar 10 kilo vandhuchu" / "Enna stock kammi?"

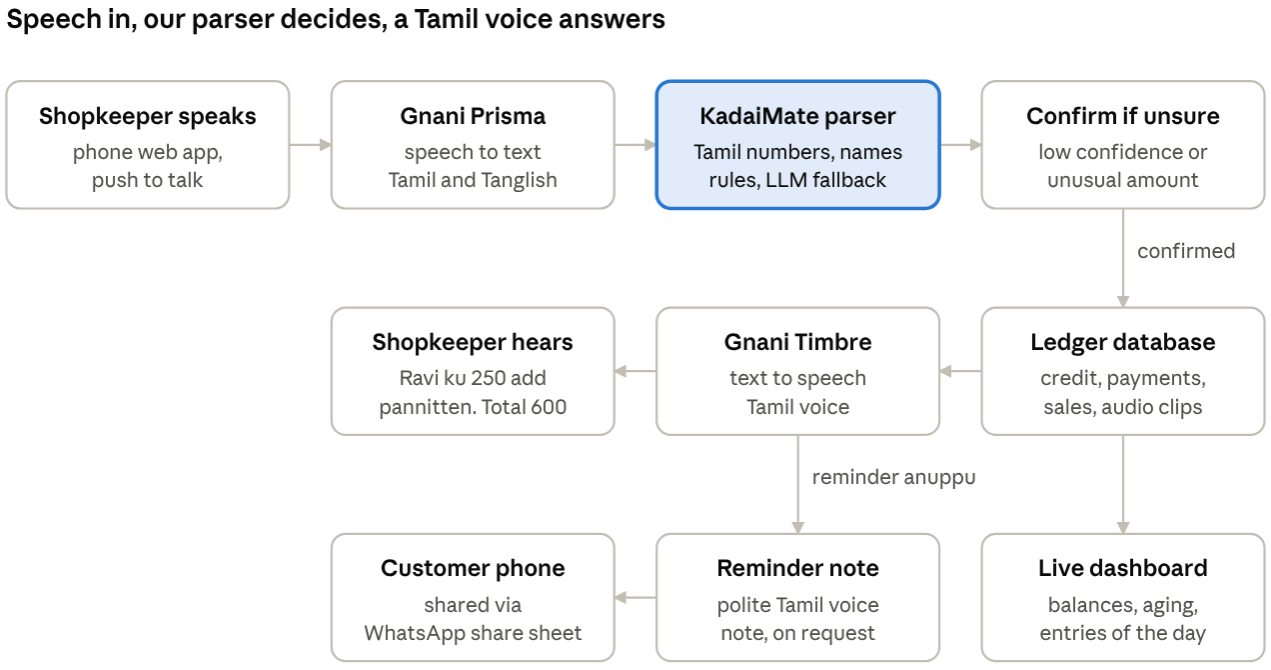
Stretch (only if ahead)

- Phone-order mode: a recorded order call (synthetic) is transcribed and logged.
- Nameless cash sales for tea and vegetable stalls.
- Excel export.

Out of scope: offline sync, PIN lock, owner and helper roles, tailor mode, supplier payables, profit estimates, PDF export. Each is useful but none shows in a 60-second video or fits 11 days.

How it works

Every command runs one loop: Prisma transcribes, our parser turns the words into a ledger action, and Timbre speaks the result.



KadaiMate pipeline · Gnani speech on both ends

The parser is our main contribution and the part to test hardest.

Proposed stack (to confirm together)

- Frontend: mobile-first web app (React or plain JS) with the browser microphone; no app-store install.
- Backend: Python FastAPI, calling Prisma and Timbre through Gnani's API.

- Parser: rules for Tamil numbers, fractions, honorifics and phonetic name matching (e.g. a Soundex-style match on the customer list); a hosted LLM returning strict JSON only when rules fail.
- Database: SQLite for the demo, one shop.
- Reminder: Timbre audio file shared through the phone's share sheet to WhatsApp; no WhatsApp Business API.
- Hosting: one small cloud instance or Render/Railway free tier.

Evaluation

We publish a real-voice accuracy table, in the same clean / noisy / phone format KiranaConnect used, so the two can be compared directly.

- **Test set:** 20 scripted commands covering every Must feature, using invented customer names and amounts.
- **Speakers:** at least 6 consenting volunteers, mixing ages, men and women, and accents (Coimbatore, Chennai, Madurai if possible), including 2 older speakers.
- **Conditions:** clean room, shop noise (real shop or played-back noise), and over a phone call.
- **Metrics per condition:** command fully correct (%), amount correct (%), name correct (%), and how often a confirmation was asked.
- **Baseline vs ours:** raw Prisma transcript parsed with simple rules, against the full KadaiMate parser. This shows what our Tamil number and name handling adds.

No real phone numbers, account details or real customer data in recordings. Recordings stay out of the public demo unless the speaker agrees.

Demo video and posting plan

The video is filmed in a real shop with a real shopkeeper who agrees to appear, and it goes live by Nov 7 so engagement has three full days to build before the cutoff.

60-second script

1. 0–08s: the paper notebook, crossed-out entries. Caption: "Every kadai in Tamil Nadu runs on this."
2. 08–25s: shop noise; the shopkeeper speaks two credit entries in Tanglish. Transcript and balance update on screen; Timbre confirms in Tamil.
3. 25–35s: a spoken correction ("205 illa 250") fixes the entry.
4. 35–50s: "Ravi ku reminder anuppu." Ravi's phone plays the polite Tamil voice note; Ravi's reaction.

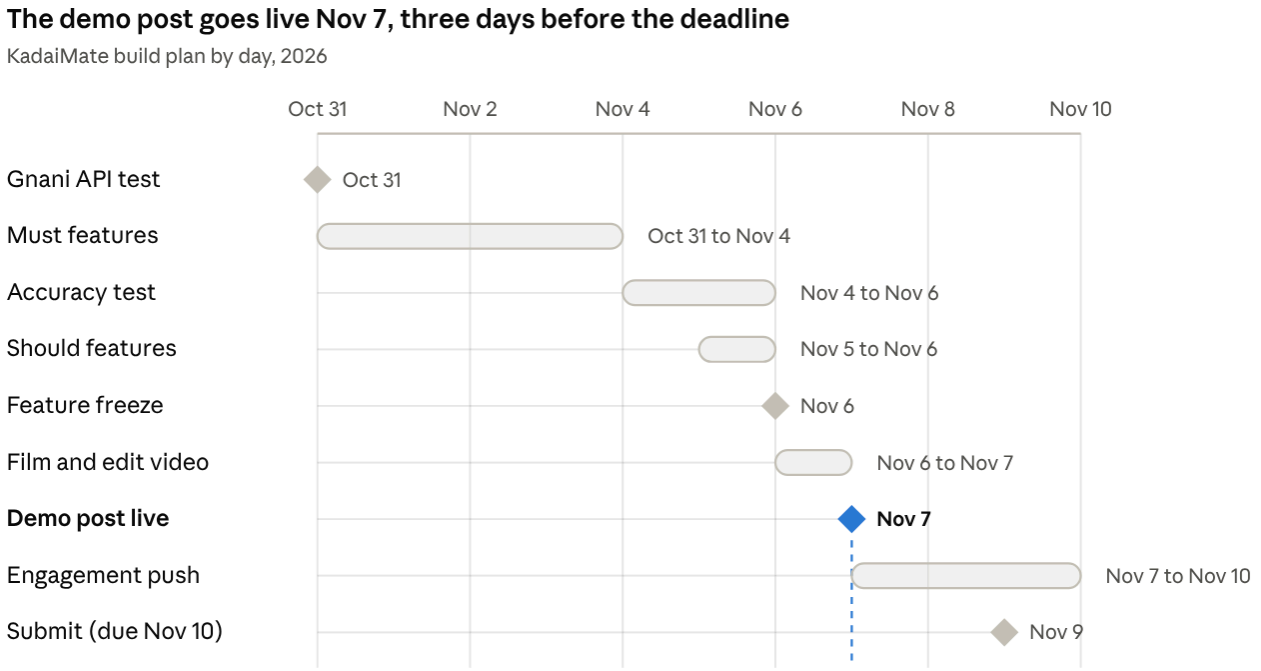
5. 50–60s: the accuracy table on screen. "Built on Gnani Prisma and Timbre." Hashtags.

Posting

- Post on both LinkedIn and X, with Tamil and English captions, tagging Gnani AI with #GnaniAI and #GreatIndianAIInternshipChallenge.
- Both teammates post or reshare; nominate one post per platform in the submission.
- Share in college, department and Tamil tech groups and ask people to comment with their own shop stories.
- No bought engagement, bots or engagement swaps: these disqualify the entry.

Timeline and team split

The post goes live on Nov 7, so engagement has three full days before the 11:59 PM IST cutoff on Nov 10; we submit on Nov 9 to leave a day of buffer.



Build plan · Oct 31 to Nov 10

Each of us is interviewed separately, so the work is split evenly: each owns one Gnani model, one hard logic problem and one screen, and both share the evaluation and launch. Names to fill in.

Area	Member A (name)	Member B (name)
Gnani model	Prisma speech to text: streaming, confidence scores	Timbre text to speech: replies and reminder voice note

Area	Member A (name)	Member B (name)
Core logic	Tanglish parser: numbers, fractions, multi-intent	Phonetic name matching, aliases, smart confirmation (unusual amount vs history)
Data	Parser output format and intent schema	Ledger data model, corrections, aging
Interface	Push-to-talk screen	Live dashboard
Evaluation	Running the accuracy tests and building the table	Test scripts and the baseline vs ours comparison
Launch	Shared: filming, write-up, posting from both accounts	Shared: filming, write-up, posting from both accounts

Risks and checks before building

The biggest risk is a Gnani model not supporting what we need in Tamil, so the first job on Oct 31 is a one-hour API test.

Verify first (Gnani Discord or docs)

- [ ] Prisma handles Tamil and Tanglish streaming, and returns a confidence score we can use.
- [ ] Timbre (in beta) has a usable Tamil voice.
- [ ] Using a non-Gnani LLM for the parsing step is allowed alongside Prisma and Timbre.
- [ ] Recording consenting volunteers for the accuracy table is allowed (the rules ban recorded calls in the project).
- [ ] How judging is regional, and whether shortlisting is still by engagement.

Risks

Risk	Fallback
Timbre has no good Tamil voice	Reply in Tanglish text on screen, Timbre in English or Hindi
Prisma mishears names or numbers	Phonetic matching plus confirmation on low confidence
LLM parsing is slow or costly	Rule-based parser for numbers and core intents, LLM only as a fallback

Risk	Fallback
5,000 credits run out	Cache test audio, run the full test set once per condition
No real shop to film	Set up a mock counter with a relative playing the shopkeeper
Scope creep	Feature freeze on Nov 6